

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE CODE: ITA0606

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO3: Bayes Theorem – Concept Learning – Maximum Likelihood – Minimum Description Length Principle – Bayes Optimal Classifier – Gibbs Algorithm – Naïve Bayes Classifier – Bayesian Belief Network – EM Algorithm – Probability Learning – Sample Complexity – Finite and Infinite Hypothesis Spaces – Mistake Bound Model, (BL 6)

Assessment Tool Weightage: 20%

QUESTIONS: 15

Total Marks:25

Annexure A

Case Study

Code of Conduct:

I S. Dharani Teja (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

1. Intelligent Hospital Disease Diagnosis System

Case Study

A hospital wants to develop an intelligent system that can diagnose diseases based on **patient symptoms and medical history**. Since different diseases can produce similar symptoms, the system must handle uncertainty and relationships between symptoms. A probabilistic model using **Bayes' Theorem, Naïve Bayes, and Bayesian Belief Networks** can be used for this purpose.

Probabilistic Diagnosis Model

The system can consider variables such as:

- Disease
- Fever
- Cough
- Headache
- Fatigue
- Age
- Previous medical history
- Laboratory results

The objective is to calculate the probability of a particular disease given the available patient information.

Bayes' Theorem provides the mathematical foundation:

$$P(D|E) = \frac{P(E|D)P(D)}{P(E)}$$

where:

- $P(D)$ = prior probability of the disease
- $P(E|D)$ = probability of observing the evidence when the patient has the disease
- $P(E)$ = overall probability of the evidence
- $P(D|E)$ = posterior probability of the disease after observing the evidence

For example, if a patient has fever and cough, the system updates the probability of diseases associated with these symptoms instead of making a fixed yes/no decision.

Role of Naïve Bayes

A Naïve Bayes classifier can classify a patient based on multiple symptoms. It assumes that the symptoms are **conditionally independent given the disease**.

For example:

```
[  
P(D|Fever,Cough)  
\propto  
P(D)P(Fever|D)P(Cough|D)  
]
```

This assumption simplifies the calculation and makes Naïve Bayes computationally efficient. However, symptoms may actually be related to one another. For example, fever and fatigue may occur together because of the same underlying disease. Therefore, the independence assumption may not always perfectly represent real medical conditions.

Role of Bayesian Belief Network

A **Bayesian Belief Network (BBN)** can represent dependencies explicitly.

For example:

Disease → Fever
Disease → Cough
Disease → Fatigue

Medical history can also influence the probability of a disease.

The advantage of a BBN is that it represents relationships between variables rather than treating all symptoms as independent. When evidence is entered into the network, the probability of related variables is updated.

Handling Uncertainty

Medical diagnosis naturally contains uncertainty because:

- The same symptom can occur in different diseases.
- Some symptoms may not be present even when a disease exists.
- Patient information may be incomplete.

- Medical test results may contain errors.

Probability allows the system to express uncertainty quantitatively. Instead of saying that a patient definitely has a disease, the system can indicate that the disease has a certain posterior probability.

Probability Learning

The system can learn probabilities from historical medical records. For example, it can estimate:

- How frequently a disease occurs.
- How frequently a symptom occurs for a particular disease.
- Relationships between symptoms and diseases.
- How medical history affects disease probability.

As more reliable patient data becomes available, these probability estimates can be improved.

Effectiveness and Real-World Constraints

The effectiveness of the system can be evaluated using measures such as **accuracy, precision, recall, and sensitivity**.

However, real-world implementation has several challenges:

1. Medical data may be incomplete or noisy.
2. Some diseases have overlapping symptoms.
3. Training data may not represent every patient population.
4. Patient privacy must be protected.
5. Incorrect probability estimates can lead to incorrect predictions.
6. The system should support doctors rather than replace professional medical judgment.

Conclusion

A combination of Bayes' Theorem, Naïve Bayes, and Bayesian Belief Networks can provide an effective framework for intelligent disease diagnosis. Bayes' Theorem updates beliefs using evidence, Naïve Bayes provides efficient classification, and BBNs represent dependencies between medical variables. The major advantage is the ability to handle uncertainty explicitly while continuously improving probability estimates from data.

2. Spam Detection System Using Naïve Bayes

Case Study

An organization wants to develop a system that automatically identifies whether an email is **spam or legitimate**. The system can use features such as suspicious words, number of links, sender information, attachments, and word frequency.

Naïve Bayes Classification

Naïve Bayes calculates the probability that an email belongs to each class.

For an email containing features $(X_1, X_2, \dots, X_n)$:

$$\begin{aligned} & [\\ & P(\text{Class} | X_1, X_2, \dots, X_n) \\ & \propto \\ & P(\text{Class}) \prod_i P(X_i | \text{Class}) \\ &] \end{aligned}$$

The classifier compares the posterior probabilities of the spam and legitimate classes.

If:

$$\begin{aligned} & [\\ & P(\text{Spam} | \text{Email}) > P(\text{Legitimate} | \text{Email}) \\ &] \end{aligned}$$

the email is classified as spam.

Bayes Optimal Classifier

The **Bayes Optimal Classifier** considers all possible hypotheses and makes a prediction based on their posterior probabilities.

Its main advantage is that it considers uncertainty over different possible models rather than selecting only one hypothesis. In theory, this can produce highly accurate decisions when the probability distribution is known correctly.

However, considering a large number of hypotheses can be computationally expensive. Naïve Bayes is therefore more practical for large-scale email classification.

Maximum Likelihood Estimation

Maximum Likelihood Estimation (MLE) estimates model parameters by selecting values that make the observed training data most probable.

For spam detection, MLE can be used to estimate probabilities such as:

[
P("offer"|Spam)
]

and

[
P("offer"|Legitimate)
]

The advantage of MLE is its simplicity and direct use of observed data. However, if the training dataset is small, the estimated probabilities may be unreliable.

Minimum Description Length

The **Minimum Description Length (MDL)** principle selects a model that minimizes the total description required to represent:

1. The model itself.
2. The data when represented using that model.

Conceptually:

[
L(Model)+L(Data|Model)
]

A model with too many unnecessary features may describe the training data extremely well but become complex and prone to overfitting. MDL attempts to find a balance between model simplicity and data representation.

MLE vs MDL

Aspect	MLE	MDL
Main goal	Maximize likelihood	Minimize total description length
Model complexity	May become high	Penalizes unnecessary complexity
Small datasets	Can be unreliable	Can help control complexity

Aspect	MLE	MDL
Generalization	Depends on model	Encourages simpler models

Effect of Sample Size

A large number of training emails provides better estimates of word probabilities. Therefore, the classifier generally becomes more reliable.

With a small dataset:

- Rare words may have unreliable probabilities.
- Some important spam patterns may not be observed.
- Classification accuracy may decrease.

With a large dataset:

- Probability estimates become more reliable.
- More spam patterns are learned.
- Generalization usually improves.

Effect of Hypothesis Space

The **hypothesis space** contains all possible models that could be considered for classification.

A larger hypothesis space provides greater flexibility but increases:

- Computational complexity.
- Required training data.
- Risk of overfitting.

A smaller hypothesis space is easier to learn but may not capture complex spam patterns.

Conclusion

Naïve Bayes provides a fast and effective solution for spam detection, while the Bayes Optimal Classifier provides a theoretically strong approach by considering uncertainty across hypotheses. MLE focuses on fitting observed data, whereas MDL balances data fitting with model simplicity. Sample size and hypothesis-space size strongly influence classification accuracy and generalization.

3. Customer Segmentation Using EM Algorithm

Case Study

A retail company wants to divide its customers into meaningful groups based on purchasing behavior. However, the actual group or cluster of each customer is unknown. Therefore, the problem contains **hidden or latent information**.

The company can use the **Expectation-Maximization (EM) algorithm** to estimate these hidden patterns and the underlying probability distributions.

Customer Data

Possible attributes include:

- Purchase frequency
- Average spending
- Number of products purchased
- Time between purchases
- Preferred product categories

The objective is to discover customer groups without having predefined labels.

Working of EM Algorithm

EM works iteratively through two major steps.

Step 1 – Expectation (E-Step)

The algorithm uses the current model parameters to calculate the probability that each customer belongs to each cluster.

For example, a customer might have:

- 70% probability of belonging to Cluster A.
- 20% probability of belonging to Cluster B.
- 10% probability of belonging to Cluster C.

These are **soft assignments**, rather than permanently assigning the customer to one group.

Step 2 – Maximization (M-Step)

Using these estimated cluster probabilities, the algorithm updates the parameters of each cluster.

The objective is to increase the likelihood of the observed customer data.

The E-step and M-step are repeated until the model converges.

Convergence

EM generally continues until:

- The change in likelihood becomes very small.
- The parameter changes become negligible.
- A predefined maximum number of iterations is reached.

The algorithm usually improves the likelihood during successive iterations, but it may converge to a **local optimum** rather than the global optimum.

Initialization Challenge

The starting values of the parameters can strongly affect the final solution.

If the initial parameters are poor, EM may converge to an undesirable local optimum.

Therefore, the company can:

- Run EM multiple times.
- Use different initial parameter values.
- Compare the final likelihood values.
- Select the best solution.

Role of Probability Learning

EM performs probability learning because it estimates both:

- The probability of cluster membership.
- The parameters describing each cluster's underlying distribution.

This is particularly useful when customer labels are unavailable.

Application in Marketing

After identifying customer segments, the company can design targeted strategies.

For example:

High-value customers: Premium offers and loyalty programs.

Occasional customers: Discount campaigns to increase purchase frequency.

Low-engagement customers: Personalized recommendations and retention campaigns.

Thus, the learned clusters can directly support business decision-making.

Limitations

The approach has several limitations:

1. It is sensitive to initialization.
2. It can converge to local optima.
3. The number of clusters may need to be selected appropriately.
4. Results depend on assumptions about the probability distribution.
5. Outliers and noisy customer data can affect the model.

Conclusion

The EM algorithm is suitable for customer segmentation when cluster membership is unknown. It uses iterative probability learning to estimate hidden memberships and model parameters. Although initialization and convergence issues must be considered, the resulting customer segments can support targeted marketing and personalized business strategies.

4. Medical Image Concept Learning Using Candidate Elimination

Case Study

A research team wants to develop a system that classifies medical images into **normal and abnormal** categories. The system receives labelled training examples and must learn a general concept that can classify previously unseen images.

Hypothesis Space

A **hypothesis space** represents all possible rules that the learning algorithm can use to classify an image.

Suppose images are described using:

- Texture
- Shape

- Intensity
- Edge characteristics

Different combinations of these attributes form different hypotheses.

For example:

Hypothesis 1:

Texture = Smooth AND Intensity = Low $\rightarrow$ Normal

Hypothesis 2:

Texture = Rough AND Shape = Irregular $\rightarrow$ Abnormal

The learning algorithm searches among such hypotheses to identify those consistent with the training examples.

Candidate Elimination

Candidate Elimination maintains two boundaries:

Specific boundary (S): The most specific hypotheses that are consistent with the training examples.

General boundary (G): The most general hypotheses that are consistent with the training examples.

The collection of hypotheses between these boundaries forms the **version space**.

Learning from Positive Examples

When a positive example is provided, the specific boundary is generalized enough to include the new example.

The general boundary is retained only if it remains consistent with the example.

Learning from Negative Examples

When a negative example is received, hypotheses that incorrectly classify the example as positive are removed or specialized.

Therefore, each labelled example reduces the number of hypotheses that remain possible.

Finite Hypothesis Space

A finite hypothesis space contains a limited number of possible hypotheses.

Advantages include:

- Easier systematic search.
- Clearer convergence behaviour.
- Easier computational analysis.
- Possibility of establishing mistake bounds.

As more labelled examples are provided, the version space becomes smaller.

Infinite Hypothesis Space

An infinite hypothesis space contains an unlimited number of possible hypotheses.

It provides greater flexibility and can represent more complex concepts. However, it creates greater challenges because:

- More training data may be required.
- Search becomes more difficult.
- Generalization becomes more dependent on inductive bias.
- Learning efficiency may decrease.

Sample Complexity

Sample complexity refers to the amount of training data required for a learner to achieve a desired level of accuracy and confidence.

If the hypothesis space is very large, more examples may be required to distinguish the correct hypothesis from competing hypotheses.

Therefore:

Larger hypothesis space → Higher potential sample complexity

Mistake Bound Model

The mistake bound model studies the number of prediction errors a learner can make before identifying a consistent concept.

For a simple finite hypothesis elimination approach, the number of possible eliminations is related to the size of the hypothesis space:

$$\begin{aligned} &[\\ \text{Mistake Bound} &\leq |H|-1 \\ &] \end{aligned}$$

where (H) represents the hypothesis space.

For example, if there are 50 possible hypotheses, a simple worst-case elimination bound is 49 mistakes.

Generalization

The ultimate objective is not simply to classify training images correctly. The learned hypothesis should also correctly classify **unseen medical images**.

A model with excessive flexibility may memorize training examples and fail on new images. A suitable hypothesis space and sufficient training data are therefore important for generalization.

Conclusion

Candidate Elimination provides a systematic approach for concept learning by maintaining the specific and general boundaries of the version space. A finite hypothesis space makes learning and mistake-bound analysis easier, whereas an infinite hypothesis space provides greater flexibility but increases learning complexity and sample requirements. Proper training data and suitable inductive bias are essential for reliable generalization.

5. Bayesian Investment Decision System

Case Study

A financial institution needs to make investment decisions under uncertain market conditions. Market behaviour depends on several factors, including economic indicators, interest rates, inflation, company performance, and market trends.

A combination of **Bayesian inference, Gibbs algorithm, and probability learning** can be used to build a system that continuously updates investment beliefs.

Bayesian Inference

Bayesian inference provides a framework for updating existing beliefs when new evidence becomes available.

The basic relationship is:

$$\begin{aligned} &[\\ &P(H|E)=\frac{P(E|H)P(H)}{P(E)} \\ &] \end{aligned}$$

where:

- (H) = investment hypothesis.
- (E) = observed market evidence.
- $(P(H))$ = prior probability.
- $(P(H|E))$ = posterior probability.

For example, the institution may initially believe that a particular investment has a certain probability of producing a positive return. When new information about market conditions arrives, this probability is updated.

Influence of Prior Knowledge

Prior knowledge can come from:

- Historical market behaviour.
- Previous investment performance.
- Expert knowledge.
- Economic indicators.
- Previous observations.

If strong prior knowledge is available, it can significantly influence the initial prediction.

However, if the prior is unrealistic, it can influence the result incorrectly, especially when only limited new data is available.

As more reliable evidence becomes available, the influence of the prior generally decreases and the observed data has greater influence.

Gibbs Algorithm

For complex Bayesian models, calculating the exact posterior probability may be difficult.

The **Gibbs algorithm** is a Markov Chain Monte Carlo sampling technique that approximates the posterior distribution.

It works by repeatedly sampling each variable from its conditional probability distribution while keeping the current values of the other variables fixed.

After many iterations, the generated samples can be used to estimate the posterior distribution.

Dynamic Probability Learning

The financial system can continuously receive new information such as:

- Changes in interest rates.
- Company financial reports.
- Market trends.
- Inflation changes.
- Economic announcements.

The probability estimates can then be updated as new evidence becomes available.

This makes the model dynamic rather than relying only on historical assumptions.

Risk Evaluation

Posterior probabilities can be used to evaluate investment risk.

For example:

- High posterior probability of positive returns → potentially favourable investment.
- High posterior probability of negative returns → higher risk.
- Similar probabilities for positive and negative outcomes → high uncertainty.

The institution can use these probabilities along with other financial measures to support investment decisions.

Computational Challenges

Bayesian models can become computationally expensive when:

- There are many variables.
- Variables have complex dependencies.
- The posterior distribution is difficult to calculate.
- A large number of Gibbs iterations is required.

Gibbs sampling may also require sufficient iterations to reach a stable distribution.

Optimization Strategies

The system can improve computational efficiency by:

1. Selecting only important variables.
2. Using appropriate initialization.
3. Monitoring convergence.
4. Removing unnecessary dependencies.
5. Using efficient sampling strategies.
6. Stopping the algorithm after sufficient convergence.

Conclusion

A Bayesian investment system can combine prior financial knowledge with continuously arriving market evidence. Bayesian inference updates investment beliefs, Gibbs sampling helps approximate complex posterior distributions, and probability learning allows the model to adapt to changing market conditions. Although computational complexity and uncertainty remain challenges, the approach provides a systematic framework for probabilistic risk assessment and decision-making.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand